Shaik Koduru Asif

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Education

Indian Institute of Technology Madras — Chennai, India
B.Tech in Civil Engineering | CGPA: 7.71/10 — *2023 – 2027*
Class XII, Board of Intermediate Education, Andhra Pradesh | 96.7% — 2023
Class X, Board of Secondary Education, Andhra Pradesh | 99% — 2021

Experience

Groop Games — May 2026 – Jul 2026
AI Engineering Intern

- Built a board-game AI in JavaScript using Minimax with alpha-beta pruning, transposition tables and move ordering
- Achieved **106.3x** median speedup and **98.3%** node reduction over naive search across 200 depth-4 benchmark positions
- Redesigned board state with BigInt bitmasks, enabling **9.4x** faster board copies and **5.3x** faster state-key generation
- Added iterative deepening with time-controlled search; scored **65.95%** against a greedy AI in **1,000** balanced matches
- Validated **1,000** positions, **2,197** penalty cases and **120** JS/Python cross-checks, confirming exact search correctness

Projects

TinyLLM: Decoder-Only Transformer Language Model — *PyTorch, RoPE, Positional Encodings*

- Implemented a **12.2M**-parameter decoder-only Transformer from scratch with dynamic masking and 256-token context
- Benchmarked learned, sinusoidal and RoPE positional encodings across context lengths ranging from 64 to 256 tokens
- Reached **9.44** perplexity with RoPE at 256-token context versus **14.51** for learned and **204.37** for sinusoidal encodings

Financial Document Intelligence RAG — *Python, BGE, BM25, CrossEncoder, ChromaDB, Streamlit*

- Architected hybrid retrieval fusing BGE embeddings and BM25 with CrossEncoder reranking for **83.3%** Recall@5
- Generated citation-grounded answers with prompt-injection guards, hitting **88.7%** faithfulness and **95.12%** relevancy
- Ingested PDF, DOCX and XLSX files with table-preserving extraction and vision-model summaries of chart images

Hierarchical Multi-Token Transformer for Ground Motion Prediction — *PyTorch, SHAP*

- Trained a Transformer on **18,500+** ESM/NGA-West seismic records to predict spectral acceleration at **111** periods
- Designed feature tokenization with a period-conditioned cross-attention decoder, attaining **0.865** R^2 beyond GMPEs
- Used SHAP and residual decomposition to verify physical consistency, lowering spectral bias by **76%** versus baseline

Image Stitching: Classical vs Deep Learning Features — *OpenCV, PyTorch, Kornia*

- Hand-coded a panorama stitching pipeline with RANSAC homography estimation, warping, blending and cropping
- Compared SIFT, ORB and DISK + LightGlue; learned matching kept **99.7%** RANSAC inliers versus **97.6%** for SIFT
- Profiled the speed trade-off: SIFT matched an image pair in **117 ms** versus **6.9 s** for DISK + LightGlue on CPU

Autonomous GitHub Issue Agent — *LangGraph, LLMs, FastAPI, PyGithub, Docker*

- Developed an 8-stage LangGraph agent that localizes bugs, writes fixes and tests, and opens PRs after human approval
- Resolved **80%** of issues with **100%** top-1 bug localization; its self-repair loop recovered **40%** of failed first patches
- Served concurrent users through a multi-tenant FastAPI backend, running LLM-written tests in an isolated sandbox

Credit Risk Assessment — *XGBoost, Scikit-learn, Pandas*

- Engineered **269** features from **145** raw columns using multi-table aggregation, domain ratios and supervised WOE
- Tuned class-weighted XGBoost with hyperparameter search to **0.771** ROC-AUC and **0.265** PR-AUC on loan defaults
- Calibrated scores with isotonic regression and cost-sensitive thresholds for **0.324** F1, **39.0%** recall and **27.7%** precision

Technical Skills

Languages & Data: Python, C++, C, SQL, JavaScript, NumPy, Pandas, PySpark, PostgreSQL, MySQL, Matplotlib
Machine Learning: PyTorch, TensorFlow, Scikit-learn, XGBoost, SHAP, OpenCV, Transformers, Computer Vision
GenAI & Tools: LangGraph, LangChain, RAG, ChromaDB, Prompt Engineering, FastAPI, Streamlit, Docker, Git
Coursework: Machine Learning Specialization (Andrew Ng), Statistics and Stochastic Processes, Stochastic Methods

Achievements

- JEE Advanced 2023: category rank **951** among **2.5 lakh** candidates (top **2.9%**); JEE Main: top **0.59%** of **11 lakh**
- ISI Admission Test: top **1–2%**, shortlisted for B.Stat (Hons.) interview; AP EAPCET: rank **606** among **2.38 lakh**
- Google PromptWars: rank **312** with a **93.13** score; invited to Ather Builders 2026 with sponsored entry and travel